# VolunTier

**DIGT2107: Practice of Software Development**
Project Iteration 3: Testing and Initial Development
Course: DIGT2107 - Fall Term 2026. Instructor: Dr. May Haidar. Due date: November 8, 2026.

## 1. Introduction

**Project Name:** VolunTier
**Team Number:** 3
**Team Members:** Jana Hussien (222143515), Mohammad Jalili (221243084), Anwar Bonunagui (222290266), Crystal Mumtaz-Shah (220409777)

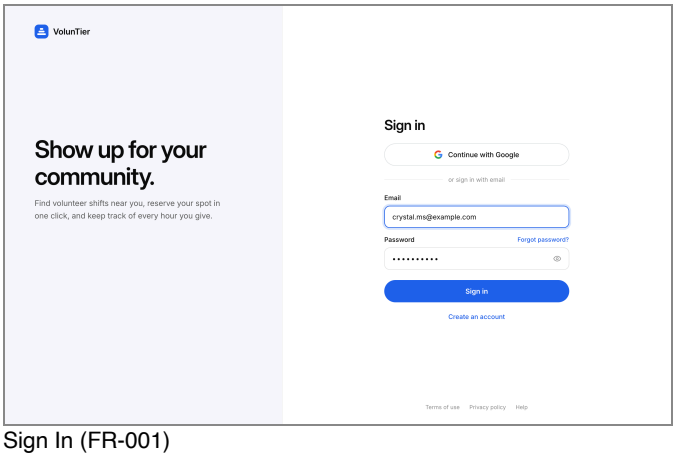
**Document Overview:** This document covers Iteration 3 of VolunTier, a Java application, built with Spring Boot, that helps community organizations post volunteer opportunities, register volunteers, record attendance and hours, and send notifications. It shows how the test cases link to the functional and non-functional requirements from Iteration 2. It also gives the repository strategy, the task allocation, and the sprint plan.

## 2. Iteration Goals and Objectives

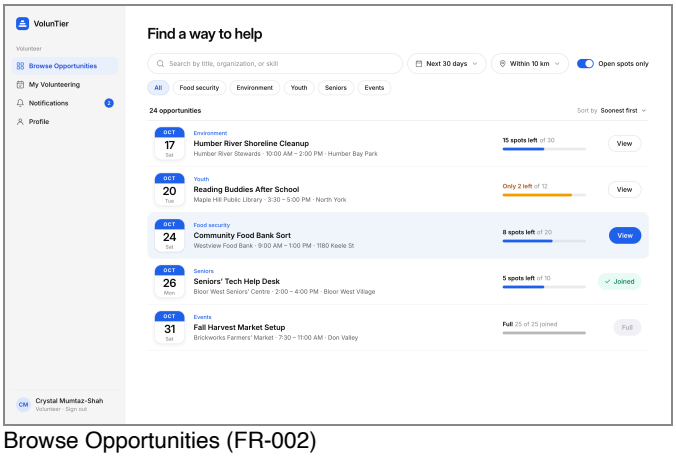
- Implement core functionality for the high-priority user stories: Role-Based Login (US1), Browse and Search Opportunities (US2), Register for an Opportunity (US3), and Manage Opportunities (US6).
- Write and run unit tests that map to specific requirements. Each test uses a test case ID in its name.
- Deliver an initial prototype that shows the link between test cases and requirements.

### Prototype Overview

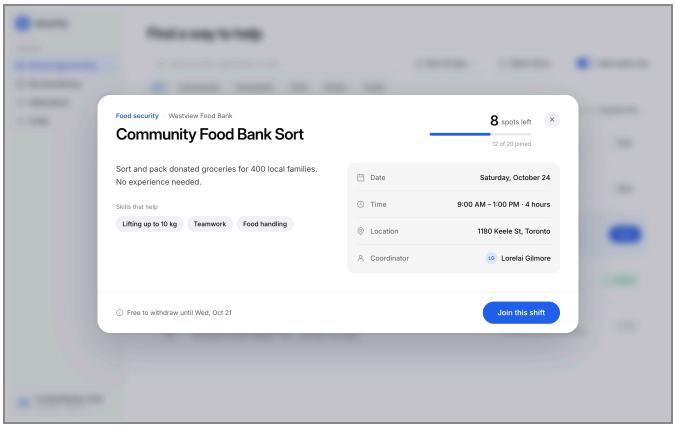
The Iteration 2 prototype has seven wireframes with mock data, and one flow diagram. The Iteration 3 prototype is a working Spring Boot application. It has a REST API and a browser page (`/`) for sign in, browse and search, register, and post an opportunity. Each panel on the page shows its requirement ID and its test case IDs. Run it with `mvn spring-boot:run` and open `http://localhost:8080`. Demo accounts: `volunteer@voluntier.test`, `coordinator@voluntier.test`, `admin@voluntier.test` (password: `password`). Data is stored in memory in this iteration.



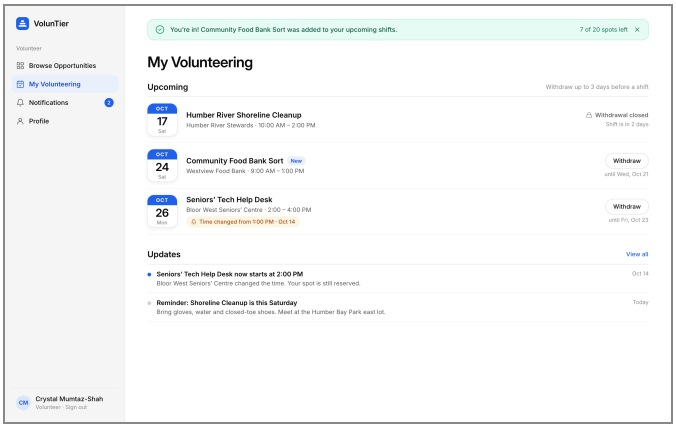
Sign In (FR-001)



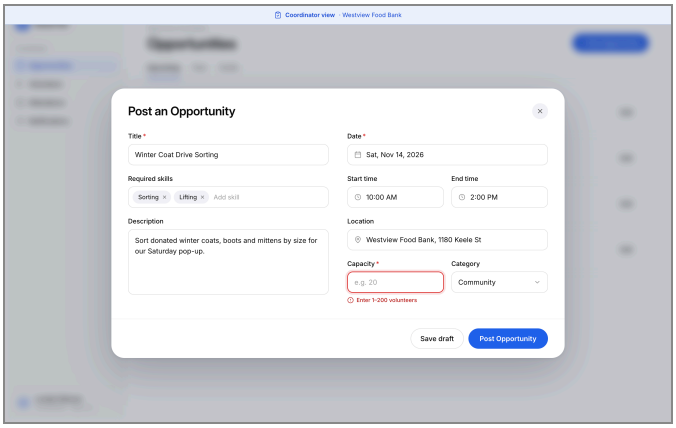
Browse Opportunities (FR-002)



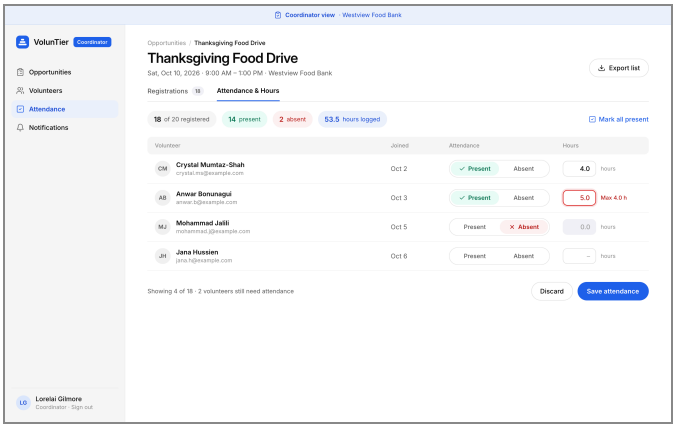

Opportunity Details and Register (FR-003)



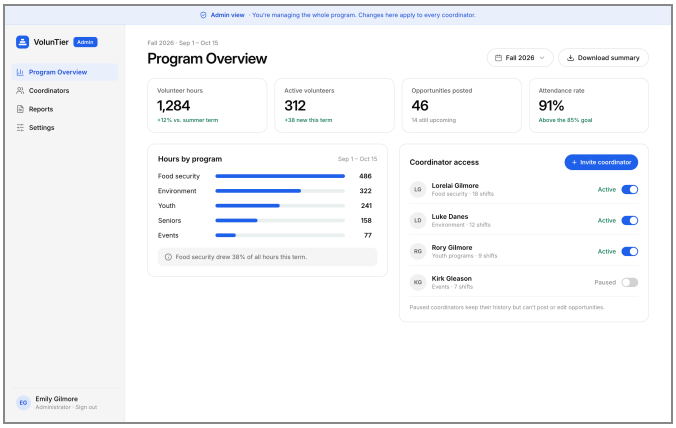

My Volunteering (FR-004, FR-005)



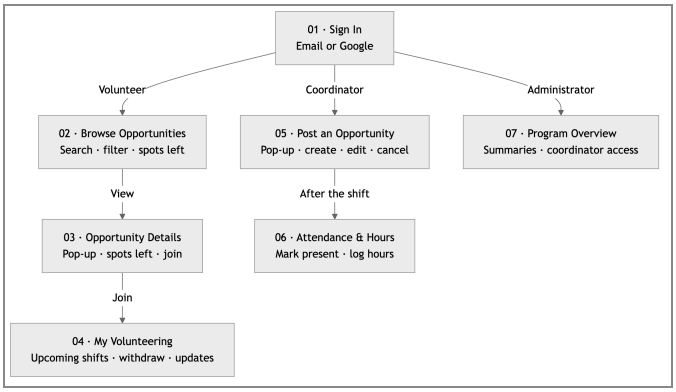

Post an Opportunity (FR-006)




Attendance and Hours (FR-007)




Program Overview (FR-008)




Screen flow

# 3. Updated Requirements and Use Cases

## 3.1 Functional Requirements

| ID | Requirement | User Story | Priority |
|---|---|---|---|
| FR-001 | The system shall allow users to log in and access features based on their assigned role: volunteer, coordinator, or administrator. | US1 | High |
| FR-002 | The system shall allow volunteers to browse, search, and filter available volunteer opportunities. | US2 | High |
| FR-003 | The system shall allow volunteers to register for available opportunities while preventing duplicate registrations and registrations beyond the capacity limit. | US3 | High |
| FR-004 | The system shall allow volunteers to withdraw from an opportunity more than three days before the event's scheduled start time. | US4 | Medium |
| FR-005 | The system shall allow volunteers to view their upcoming registrations, participation history, and completed volunteer hours. | US5 | Medium |
| FR-006 | The system shall allow volunteer coordinators to create, post, edit, and cancel volunteer opportunities. | US6 | High |
| FR-007 | The system shall allow volunteer coordinators to view registrations and record volunteer attendance and completed hours. | US7 | Medium |
| FR-008 | The system shall allow organization administrators to manage coordinator access and review program summaries. | US8 | Medium |
| FR-009 | The system shall send registered volunteers an in-app or email notification when an opportunity's date, time, location, or status changes. | US9 | Medium |

IDs FR-001 to FR-009 match requirements 1 to 9 in the Iteration 2 document.

## 3.2 Non-Functional Requirements

| ID | Quality | Requirement |
|---|---|---|
| NFR-001 | Performance | Opportunity lists, registration results, and participation records appear within two seconds under normal use. |
| NFR-002 | Usability | The Java desktop interface uses clear labels, readable text, consistent navigation, and understandable error messages. |
| NFR-003 | Reliability | The system saves registration, withdrawal, attendance, and volunteer-hour records accurately without duplicate records. |
| NFR-004 | Security | The system restricts features by user role (volunteer, coordinator, administrator). |
| NFR-005 | Scalability | The database supports hundreds of users and thousands of participation records. |
| NFR-006 | Validation | The system validates required fields, dates, capacity limits, and volunteer hours before saving. |
| NFR-007 | Maintainability | The application uses a modular structure. Interface, business rules, and data storage can change separately. |

**Change in Iteration 3:** the team now builds the interface with Spring Boot and a browser page. The Iteration 2 text named a Java desktop interface. NFR-002 (Usability) applies to the browser interface.

## 3.3 Use Cases

| ID | Use Case | Actor | Requirement |
|---|---|---|---|
| UC-01 | Browse Available Opportunities | Volunteer | FR-002 |
| UC-02 | Search and Filter Opportunities | Volunteer | FR-002 |
| UC-03 | Register for an Opportunity | Volunteer | FR-003 |
| UC-04 | Withdraw from an Opportunity | Volunteer | FR-004 |

| | | | |
|---|---|---|---|
| UC-05 | View Participation History | Volunteer | FR-005 |
| UC-06 | Manage and Post Opportunities | Coordinator | FR-006, FR-009 |
| UC-07 | View Volunteer Registrations | Coordinator | FR-007 |
| UC-08 | Record Attendance and Hours | Coordinator | FR-007 |
| UC-09 | Manage Coordinator Access | Administrator | FR-008 |
| UC-10 | Review Program Summaries | Administrator | FR-008 |

The full use case descriptions (main flow, alternative flow, postconditions) are in the Iteration 2 document. No use case changed in Iteration 3.

# 4. Requirement to Test Case Traceability

## 4.1 Traceability Matrix: Functional Requirements

| Req. ID | Requirement Description | Test Case ID | Test Case Description | Coverage |
|---|---|---|---|---|
| FR-001 | The system shall allow users to log in and access features based on their assigned role: volunteer, coordinator, or administrator. | TC-001, TC-002, TC-003, TC-004, TC-005, TC-006, TC-007 | TC-001: Login - Valid Volunteer Credentials<br>TC-002: Login - Valid Coordinator Credentials<br>TC-003: Login - Valid Administrator Credentials<br>TC-004: Login - Invalid Password<br>TC-005: Login - Empty Required Fields<br>TC-006: Role Restriction - Volunteer Cannot Access Coordinator Features<br>TC-007: Role Restriction - Coordinator Cannot Access Administrator Features | Fully Covered |
| FR-002 | The system shall allow volunteers to browse, search, and filter available volunteer opportunities. | TC-008, TC-009, TC-010, TC-011, TC-012 | TC-008: Browse Opportunities - List Displays Details<br>TC-009: Search Opportunities - Matching Term<br>TC-010: Search Opportunities - No Matches<br>TC-011: Filter Opportunities - By Category or Date<br>TC-012: View Opportunity Details | Fully Covered |
| FR-003 | The system shall allow volunteers to register for available opportunities while preventing duplicate registrations and registrations beyond the capacity limit. | TC-013, TC-014, TC-015, TC-016 | TC-013: Register for Opportunity - Space Available<br>TC-014: Register for Opportunity - Duplicate Registration<br>TC-015: Register for Opportunity - Capacity Reached<br>TC-016: Register for Opportunity - Last Remaining Space | Fully Covered |
| FR-004 | The system shall allow volunteers to withdraw from an opportunity more than three days before the event's scheduled start time. | TC-017, TC-018, TC-019 | TC-017: Withdraw - More Than Three Days Before Event<br>TC-018: Withdraw - Within Three Days of Event<br>TC-019: Withdraw - Exactly Three Days Boundary | Fully Covered |
| FR-005 | The system shall allow volunteers to view their upcoming registrations, participation history, and completed volunteer hours. | TC-020, TC-021, TC-022 | TC-020: My Volunteering - Upcoming and History<br>TC-021: My Volunteering - Hours Show After Coordinator Records Them<br>TC-022: My Volunteering - No Records | Fully Covered |

| | | | | |
|---|---|---|---|---|
| FR-006 | The system shall allow volunteer coordinators to create, post, edit, and cancel volunteer opportunities. | TC-023, TC-024, TC-025, TC-026, TC-027 | TC-023: Create Opportunity - Valid Inputs<br>TC-024: Create Opportunity - Missing Required Fields<br>TC-025: Create Opportunity - Invalid Date or Capacity<br>TC-026: Edit Opportunity - Valid Change<br>TC-027: Cancel Opportunity - Registered Volunteers Notified | Partially Covered |
| FR-007 | The system shall allow volunteer coordinators to view registrations and record volunteer attendance and completed hours. | TC-021, TC-028, TC-029, TC-030 | TC-021: My Volunteering - Hours Show After Coordinator Records Them<br>TC-028: View Registrations for an Opportunity<br>TC-029: Record Attendance and Hours - Valid<br>TC-030: Record Hours - Invalid Hours | Partially Covered |
| FR-008 | The system shall allow organization administrators to manage coordinator access and review program summaries. | TC-031, TC-032, TC-033, TC-034 | TC-031: Grant Coordinator Access<br>TC-032: Remove Coordinator Access<br>TC-033: Review Program Summaries<br>TC-034: Program Summaries - No Data For Period | Partially Covered |
| FR-009 | The system shall send registered volunteers an in-app or email notification when an opportunity's date, time, location, or status changes. | TC-027, TC-035, TC-036 | TC-027: Cancel Opportunity - Registered Volunteers Notified<br>TC-035: Notification - Opportunity Time or Location Changed<br>TC-036: Notification - Unregistered Volunteer Not Notified | Partially Covered |

Coverage shows the planned coverage from test design. The team must confirm it after the tests run.

### 4.2 Traceability Matrix: Non-Functional Requirements

| Req. ID | Requirement Description | Test Case ID | Test Case Description |
|---|---|---|---|
| NFR-001 | Performance: Opportunity lists, registration results, and participation records appear within two seconds under normal use. | TC-037 | TC-037: Performance - Lists Load Within Two Seconds |
| NFR-002 | Usability: The Java desktop interface uses clear labels, readable text, consistent navigation, and understandable error messages. | None | No test case yet |
| NFR-003 | Reliability: The system saves registration, withdrawal, attendance, and volunteer-hour records accurately without duplicate records. | TC-038 | TC-038: Reliability - No Duplicate Records On Rapid Double Click |
| NFR-004 | Security: The system restricts features by user role (volunteer, coordinator, administrator). | TC-006, TC-007 | TC-006: Role Restriction - Volunteer Cannot Access Coordinator Features<br>TC-007: Role Restriction - Coordinator Cannot Access Administrator Features |
| NFR-005 | Scalability: The database supports hundreds of users and thousands of participation records. | None | No test case yet |

| | | | |
|---|---|---|---|
| NFR-006 | Validation: The system validates required fields, dates, capacity limits, and volunteer hours before saving. | TC-005, TC-024, TC-025, TC-030 | TC-005: Login - Empty Required Fields<br>TC-024: Create Opportunity - Missing Required Fields<br>TC-025: Create Opportunity - Invalid Date or Capacity<br>TC-030: Record Hours - Invalid Hours |
| NFR-007 | Maintainability: The application uses a modular structure. Interface, business rules, and data storage can change separately. | None | No test case yet |

Coverage status is not required for non-functional requirements.

### 4.3 Explanation

**How the test cases validate each requirement.** Each functional requirement has at least two test cases. Each has one test for the normal path and one test for a rejected input or a blocked action. Registration (FR-003) and withdrawal (FR-004) also have boundary tests for the last free space and the exact three-day cutoff.

**Gaps.**

- FR-006: Add tests for invalid input on edit, and for a coordinator who edits an opportunity that another coordinator owns.
- FR-007: Add tests for attendance on an event that has not yet taken place, and for hours above the event length.
- FR-008: Add a test that the summary totals equal the stored records.
- FR-009: Email delivery is untested. Add a test for each changed field (date, time, location, status).
- NFR-002 (Usability), NFR-005 (Scalability), and NFR-007 (Maintainability) have no test case. Usability is checked by team review of the screens. Scalability needs a data load script. Maintainability is checked by code review of the module structure.

**Coverage status.** 5 functional requirements are fully covered. 4 are partially covered. None are uncovered. The team will update this status after the tests run.

# 5. Test Plan and Results

## 5.1 Test Plan

- **Scope:** all functional requirements FR-001 to FR-009. Priority goes to FR-001, FR-002, FR-003, and FR-006 (high-priority stories).
- **Unit tests:** JUnit 5 tests for business rules. These rules are role checks, duplicate and capacity checks, the three-day withdrawal rule, and input validation. Test names include the test case ID, for example `TC014_duplicateRegistrationIsRejected`.
- **Integration tests:** tests of the data storage layer for registration, withdrawal, attendance, and hours records. These tests check that no duplicate record is created.
- **API tests:** Spring MockMvc tests start the application and call the REST API (login, browse, search, register, post).
- **Manual tests:** the team runs the remaining screen-level test cases in section 6 on the browser page.
- **Environment:** Java 17, Spring Boot 3.3, JUnit 5, Spring MockMvc, in-memory data store, test data with sample data (hundreds of users, thousands of records for the performance test).
- **Entry criteria:** the feature is merged to the `development` branch and builds without error.
- **Exit criteria:** all High priority test cases pass. No High priority defect stays open.
- **Risk:** the team can run out of time for the notification feature (FR-009). The sprint plan delays this feature to Week 2 so it does not block the high-priority stories.

## 5.2 Results

The team has 38 test cases (15 High, 19 Medium, 4 Low). The command `mvn test` runs 34 automated tests. All 34 pass (27 unit tests and 7 API tests). They cover 24 of the 38 test cases. The other test cases need features that the team builds in Week 2: attendance and hours, admin features, notifications, My Volunteering, and the performance test. TC-027 is only partly run: the cancel rule passes (TC027a), but the volunteer notification does not exist yet.

| Priority | Test cases | Run | Pass | Fail | Not run |
|---|---|---|---|---|---|
| High | 15 | 14 | 14 | 0 | 1 |
| Medium | 19 | 9 | 9 | 0 | 10 |
| Low | 4 | 1 | 1 | 0 | 3 |
| Total | 38 | 24 | 24 | 0 | 14 |

Test cases run and passed: TC-001, TC-002, TC-003, TC-004, TC-005, TC-006, TC-007, TC-008, TC-009, TC-010, TC-011, TC-012, TC-013, TC-014, TC-015, TC-016, TC-017, TC-018, TC-019, TC-023, TC-024, TC-025, TC-026, TC-038.

## 6. Detailed Test Case Descriptions

### TC-001

| | |
|---|---|
| **Test Case ID** | TC-001 |
| **Related Requirement(s)** | FR-001 |
| **Title** | Login - Valid Volunteer Credentials |
| **Preconditions** | A volunteer account exists. The user is on the Sign In screen. |
| **Test Steps** | 1. Enter the volunteer email and password<br>2. Click "Sign In" |
| **Expected Result** | The volunteer dashboard opens. Only volunteer features show. |
| **Actual Result** | As expected. Automated test TC001 passed. |
| **Status** | Pass |
| **Priority** | High |

### TC-002

| | |
|---|---|
| **Test Case ID** | TC-002 |
| **Related Requirement(s)** | FR-001 |
| **Title** | Login - Valid Coordinator Credentials |
| **Preconditions** | A coordinator account exists. The user is on the Sign In screen. |
| **Test Steps** | 1. Enter the coordinator email and password<br>2. Click "Sign In" |
| **Expected Result** | The coordinator dashboard opens with opportunity management features. |
| **Actual Result** | As expected. Automated test TC002 passed. |
| **Status** | Pass |
| **Priority** | High |

### TC-003

| | |
|---|---|
| **Test Case ID** | TC-003 |
| **Related Requirement(s)** | FR-001 |
| **Title** | Login - Valid Administrator Credentials |
| **Preconditions** | An administrator account exists. The user is on the Sign In screen. |
| **Test Steps** | 1. Enter the administrator email and password<br>2. Click "Sign In" |
| **Expected Result** | The administrator dashboard opens with user management and program summaries. |
| **Actual Result** | As expected. Automated test TC003 passed. |
| **Status** | Pass |
| **Priority** | High |

### TC-004

| | |
|---|---|
| **Test Case ID** | TC-004 |
| **Related Requirement(s)** | FR-001 |
| **Title** | Login - Invalid Password |
| **Preconditions** | A volunteer account exists. The user is on the Sign In screen. |
| **Test Steps** | 1. Enter a valid email and a wrong password<br>2. Click "Sign In" |
| **Expected Result** | Login fails. A clear error message shows. The user stays on the Sign In screen. |
| **Actual Result** | As expected. Automated test TC004 passed. |
| **Status** | Pass |
| **Priority** | High |

### TC-005

| | |
|---|---|
| **Test Case ID** | TC-005 |
| **Related Requirement(s)** | FR-001, NFR-006 |
| **Title** | Login - Empty Required Fields |
| **Preconditions** | The user is on the Sign In screen. |
| **Test Steps** | 1. Leave email and password empty<br>2. Click "Sign In" |
| **Expected Result** | The system marks the required fields and shows an error. No login occurs. |
| **Actual Result** | As expected. Automated test TC005 passed. |
| **Status** | Pass |
| **Priority** | Medium |

### TC-006

| | |
|---|---|
| **Test Case ID** | TC-006 |
| **Related Requirement(s)** | FR-001, NFR-004 |
| **Title** | Role Restriction - Volunteer Cannot Access Coordinator Features |
| **Preconditions** | The user is logged in as a Volunteer. |
| **Test Steps** | 1. Look for "Post an Opportunity" and "Attendance & Hours" in the navigation<br>2. Try to open the coordinator page directly (URL or menu) |
| **Expected Result** | The features are hidden or access is denied. No coordinator data shows. |
| **Actual Result** | As expected. Automated test TC006 passed. |
| **Status** | Pass |
| **Priority** | High |

### TC-007

| | |
|---|---|
| **Test Case ID** | TC-007 |
| **Related Requirement(s)** | FR-001, NFR-004 |
| **Title** | Role Restriction - Coordinator Cannot Access Administrator Features |
| **Preconditions** | The user is logged in as a Coordinator. |
| **Test Steps** | 1. Look for "Manage Coordinator Access" and "Program Overview" in the navigation<br>2. Try to open the administrator page directly |
| **Expected Result** | Access is denied. No administrator data shows. |
| **Actual Result** | As expected. Automated test TC007 passed. |
| **Status** | Pass |
| **Priority** | High |

### TC-008

| | |
|---|---|
| **Test Case ID** | TC-008 |
| **Related Requirement(s)** | FR-002 |
| **Title** | Browse Opportunities - List Displays Details |
| **Preconditions** | The user is logged in as a Volunteer. At least three opportunities exist. |
| **Test Steps** | 1. Open the Browse Opportunities page |
| **Expected Result** | Each opportunity shows date, location, description, and remaining spaces. |
| **Actual Result** | As expected. Automated test TC008 passed. |
| **Status** | Pass |
| **Priority** | High |

### TC-009

| | |
|---|---|
| **Test Case ID** | TC-009 |
| **Related Requirement(s)** | FR-002 |
| **Title** | Search Opportunities - Matching Term |
| **Preconditions** | The user is logged in as a Volunteer. An opportunity titled "Food Bank Sorting" exists. |
| **Test Steps** | 1. Open Browse Opportunities<br>2. Enter "Food Bank" in the search field |
| **Expected Result** | Only matching opportunities show, including "Food Bank Sorting". |
| **Actual Result** | As expected. Automated test TC009 passed. |
| **Status** | Pass |
| **Priority** | High |

### TC-010

| | |
|---|---|
| **Test Case ID** | TC-010 |
| **Related Requirement(s)** | FR-002 |
| **Title** | Search Opportunities - No Matches |
| **Preconditions** | The user is logged in as a Volunteer. |
| **Test Steps** | 1. Open Browse Opportunities<br>2. Enter a term that matches nothing (e.g., "zzzzqq") |
| **Expected Result** | A "no matches" message shows. The user can change the search or filters. |
| **Actual Result** | As expected. Automated test TC010 passed. |
| **Status** | Pass |
| **Priority** | Medium |

### TC-011

| | |
|---|---|
| **Test Case ID** | TC-011 |
| **Related Requirement(s)** | FR-002 |
| **Title** | Filter Opportunities - By Category or Date |
| **Preconditions** | The user is logged in as a Volunteer. Opportunities exist in more than one category or date. |
| **Test Steps** | 1. Open Browse Opportunities<br>2. Select a filter value |
| **Expected Result** | The list shows only opportunities that match the filter. |
| **Actual Result** | As expected. Automated test TC011 passed. |
| **Status** | Pass |
| **Priority** | Medium |

### TC-012

| | |
|---|---|
| **Test Case ID** | TC-012 |
| **Related Requirement(s)** | FR-002 |
| **Title** | View Opportunity Details |
| **Preconditions** | The user is logged in as a Volunteer. Opportunities exist. |
| **Test Steps** | 1. Open Browse Opportunities<br>2. Select one opportunity |
| **Expected Result** | The Opportunity Details page shows title, description, date, time, location, capacity, and a Register button. |
| **Actual Result** | As expected. Automated test TC012 passed. |
| **Status** | Pass |
| **Priority** | Medium |

### TC-013

| | |
|---|---|
| **Test Case ID** | TC-013 |
| **Related Requirement(s)** | FR-003 |
| **Title** | Register for Opportunity - Space Available |
| **Preconditions** | Logged in as a Volunteer. The opportunity has open spaces. The volunteer is not registered. |
| **Test Steps** | 1. Open the opportunity details<br>2. Click "Register" |
| **Expected Result** | A confirmation shows. The opportunity appears in Upcoming Registrations. Remaining spaces fall by one. |
| **Actual Result** | As expected. Automated test TC013 passed. |
| **Status** | Pass |
| **Priority** | High |

### TC-014

| | |
|---|---|
| **Test Case ID** | TC-014 |
| **Related Requirement(s)** | FR-003 |
| **Title** | Register for Opportunity - Duplicate Registration |
| **Preconditions** | Logged in as a Volunteer. The volunteer is already registered for the opportunity. |
| **Test Steps** | 1. Open the same opportunity details<br>2. Click "Register" |
| **Expected Result** | The system refuses and explains that the volunteer is already registered. No second record is created. |
| **Actual Result** | As expected. Automated test TC014 passed. |
| **Status** | Pass |
| **Priority** | High |

### TC-015

| | |
|---|---|
| **Test Case ID** | TC-015 |
| **Related Requirement(s)** | FR-003 |
| **Title** | Register for Opportunity - Capacity Reached |
| **Preconditions** | Logged in as a Volunteer. The opportunity has 0 remaining spaces. |
| **Test Steps** | 1. Open the opportunity details<br>2. Click "Register" |
| **Expected Result** | The system refuses and explains that the opportunity is full. No registration is saved. |
| **Actual Result** | As expected. Automated test TC015 passed. |
| **Status** | Pass |
| **Priority** | High |

### TC-016

| | |
|---|---|
| **Test Case ID** | TC-016 |
| **Related Requirement(s)** | FR-003 |
| **Title** | Register for Opportunity - Last Remaining Space |
| **Preconditions** | Logged in as a Volunteer. The opportunity has exactly 1 remaining space. |
| **Test Steps** | 1. Register for the opportunity<br>2. Log in as a second volunteer and try to register |
| **Expected Result** | The first volunteer registers. Remaining spaces show 0. The second volunteer is refused. |
| **Actual Result** | As expected. Automated test TC016 passed. |
| **Status** | Pass |
| **Priority** | Medium |

### TC-017

| | |
|---|---|
| **Test Case ID** | TC-017 |
| **Related Requirement(s)** | FR-004 |
| **Title** | Withdraw - More Than Three Days Before Event |
| **Preconditions** | Logged in as a Volunteer. The volunteer is registered for an event 7 days away. |
| **Test Steps** | 1. Open My Volunteering<br>2. Select the opportunity and click "Withdraw"<br>3. Confirm |
| **Expected Result** | The registration is removed. A confirmation shows. The space is available again. |
| **Actual Result** | As expected. Automated test TC017 passed. |
| **Status** | Pass |
| **Priority** | Medium |

### TC-018

| | |
|---|---|
| **Test Case ID** | TC-018 |
| **Related Requirement(s)** | FR-004 |
| **Title** | Withdraw - Within Three Days of Event |
| **Preconditions** | Logged in as a Volunteer. The volunteer is registered for an event 2 days away. |
| **Test Steps** | 1. Open My Volunteering<br>2. Select the opportunity and click "Withdraw" |
| **Expected Result** | Withdrawal is blocked. The message explains the three-day cutoff. The registration stays. |
| **Actual Result** | As expected. Automated test TC018 passed. |
| **Status** | Pass |
| **Priority** | Medium |

### TC-019

| | |
|---|---|
| **Test Case ID** | TC-019 |
| **Related Requirement(s)** | FR-004 |
| **Title** | Withdraw - Exactly Three Days Boundary |
| **Preconditions** | Logged in as a Volunteer. The volunteer is registered for an event exactly 72 hours away. |
| **Test Steps** | 1. Open My Volunteering<br>2. Select the opportunity and click "Withdraw" |
| **Expected Result** | Withdrawal is blocked because the event is not more than three days away. |
| **Actual Result** | As expected. Automated test TC019 passed. |
| **Status** | Pass |
| **Priority** | Low |

### TC-020

| | |
|---|---|
| **Test Case ID** | TC-020 |
| **Related Requirement(s)** | FR-005 |
| **Title** | My Volunteering - Upcoming and History |
| **Preconditions** | Logged in as a Volunteer. The volunteer has upcoming and past registrations. |
| **Test Steps** | 1. Open My Volunteering |
| **Expected Result** | Upcoming registrations and past participation show in separate, clear sections. |
| **Actual Result** | (To be filled after test execution) |
| **Status** | (Pass/Fail/Blocked) |
| **Priority** | Medium |

### TC-021

| | |
|---|---|
| **Test Case ID** | TC-021 |
| **Related Requirement(s)** | FR-005, FR-007 |
| **Title** | My Volunteering - Hours Show After Coordinator Records Them |
| **Preconditions** | A coordinator recorded 3 hours for the volunteer. Logged in as that Volunteer. |
| **Test Steps** | 1. Open My Volunteering<br>2. View participation history |
| **Expected Result** | The history shows the opportunity as attended with 3 completed hours. The total hours include 3. |
| **Actual Result** | (To be filled after test execution) |
| **Status** | (Pass/Fail/Blocked) |
| **Priority** | Medium |

### TC-022

| | |
|---|---|
| **Test Case ID** | TC-022 |
| **Related Requirement(s)** | FR-005 |
| **Title** | My Volunteering - No Records |
| **Preconditions** | Logged in as a new Volunteer with no registrations. |
| **Test Steps** | 1. Open My Volunteering |
| **Expected Result** | A helpful message shows that there are no records. |
| **Actual Result** | (To be filled after test execution) |
| **Status** | (Pass/Fail/Blocked) |
| **Priority** | Low |

### TC-023

| | |
|---|---|
| **Test Case ID** | TC-023 |
| **Related Requirement(s)** | FR-006 |
| **Title** | Create Opportunity - Valid Inputs |
| **Preconditions** | Logged in as a Coordinator. |
| **Test Steps** | 1. Open Post an Opportunity<br>2. Enter title, description, future date, time, location, and capacity 10<br>3. Click "Post" |
| **Expected Result** | The opportunity saves. It appears in the list that volunteers see. |
| **Actual Result** | As expected. Automated test TC023 passed. |
| **Status** | Pass |
| **Priority** | High |

### TC-024

| | |
|---|---|
| **Test Case ID** | TC-024 |
| **Related Requirement(s)** | FR-006, NFR-006 |
| **Title** | Create Opportunity - Missing Required Fields |
| **Preconditions** | Logged in as a Coordinator. |
| **Test Steps** | 1. Open Post an Opportunity<br>2. Leave the title empty and fill other fields<br>3. Click "Post" |
| **Expected Result** | The system explains the missing field. Nothing is saved. |
| **Actual Result** | As expected. Automated test TC024 passed. |
| **Status** | Pass |
| **Priority** | High |

### TC-025

| | |
|---|---|
| **Test Case ID** | TC-025 |
| **Related Requirement(s)** | FR-006, NFR-006 |
| **Title** | Create Opportunity - Invalid Date or Capacity |
| **Preconditions** | Logged in as a Coordinator. |
| **Test Steps** | 1. Open Post an Opportunity<br>2. Enter a past date or a capacity of 0 or a negative number<br>3. Click "Post" |
| **Expected Result** | The system rejects the input with a clear message. Nothing is saved. |
| **Actual Result** | As expected. Automated test TC025 passed. |
| **Status** | Pass |
| **Priority** | Medium |

### TC-026

| | |
|---|---|
| **Test Case ID** | TC-026 |
| **Related Requirement(s)** | FR-006 |
| **Title** | Edit Opportunity - Valid Change |
| **Preconditions** | Logged in as a Coordinator. An opportunity exists. |
| **Test Steps** | 1. Open the opportunity for editing<br>2. Change the description<br>3. Click "Save" |
| **Expected Result** | The change saves and shows in the opportunity list. |
| **Actual Result** | As expected. Automated test TC026 passed. |
| **Status** | Pass |
| **Priority** | Medium |

### TC-027

| | |
|---|---|
| **Test Case ID** | TC-027 |
| **Related Requirement(s)** | FR-006, FR-009 |
| **Title** | Cancel Opportunity - Registered Volunteers Notified |
| **Preconditions** | Logged in as a Coordinator. An opportunity has two registered volunteers. |
| **Test Steps** | 1. Open the opportunity<br>2. Click "Cancel Opportunity" and confirm |
| **Expected Result** | The status changes to cancelled. Both volunteers receive a notification that names the opportunity. |
| **Actual Result** | (To be filled after test execution) |
| **Status** | (Pass/Fail/Blocked) |
| **Priority** | High |

### TC-028

| | |
|---|---|
| **Test Case ID** | TC-028 |
| **Related Requirement(s)** | FR-007 |
| **Title** | View Registrations for an Opportunity |
| **Preconditions** | Logged in as a Coordinator. An opportunity has registrations. |
| **Test Steps** | 1. Select the opportunity<br>2. Open its registration list |
| **Expected Result** | The registered volunteers and the number of available places show. No data changes. |
| **Actual Result** | (To be filled after test execution) |
| **Status** | (Pass/Fail/Blocked) |
| **Priority** | Medium |

### TC-029

| | |
|---|---|
| **Test Case ID** | TC-029 |
| **Related Requirement(s)** | FR-007 |
| **Title** | Record Attendance and Hours - Valid |
| **Preconditions** | Logged in as a Coordinator. The event date has passed. Volunteers are registered. |
| **Test Steps** | 1. Open the registration list<br>2. Mark one volunteer as attended<br>3. Enter 3 hours<br>4. Click "Save" |
| **Expected Result** | Attendance and hours save. They appear in the volunteer's participation history. |
| **Actual Result** | (To be filled after test execution) |
| **Status** | (Pass/Fail/Blocked) |
| **Priority** | Medium |

### TC-030

| | |
|---|---|
| **Test Case ID** | TC-030 |
| **Related Requirement(s)** | FR-007, NFR-006 |
| **Title** | Record Hours - Invalid Hours |
| **Preconditions** | Logged in as a Coordinator. The event date has passed. |
| **Test Steps** | 1. Open the registration list<br>2. Mark a volunteer as attended<br>3. Enter -2, or a letter, or leave empty<br>4. Click "Save" |
| **Expected Result** | The system asks for a correction. Nothing is saved. |
| **Actual Result** | (To be filled after test execution) |
| **Status** | (Pass/Fail/Blocked) |
| **Priority** | Medium |

### TC-031

| | |
|---|---|
| **Test Case ID** | TC-031 |
| **Related Requirement(s)** | FR-008 |
| **Title** | Grant Coordinator Access |
| **Preconditions** | Logged in as an Administrator. A volunteer account exists. |
| **Test Steps** | 1. Open the user management page<br>2. Select the volunteer<br>3. Grant coordinator access and save |
| **Expected Result** | The change saves. After the next login, the user sees coordinator features. |
| **Actual Result** | (To be filled after test execution) |
| **Status** | (Pass/Fail/Blocked) |
| **Priority** | Medium |

### TC-032

| | |
|---|---|
| **Test Case ID** | TC-032 |
| **Related Requirement(s)** | FR-008 |
| **Title** | Remove Coordinator Access |
| **Preconditions** | Logged in as an Administrator. A coordinator account exists. |
| **Test Steps** | 1. Open the user management page<br>2. Select the coordinator<br>3. Remove coordinator access and save |
| **Expected Result** | The change saves. The user no longer sees coordinator features. |
| **Actual Result** | (To be filled after test execution) |
| **Status** | (Pass/Fail/Blocked) |
| **Priority** | Medium |

### TC-033

| | |
|---|---|
| **Test Case ID** | TC-033 |
| **Related Requirement(s)** | FR-008 |
| **Title** | Review Program Summaries |
| **Preconditions** | Logged in as an Administrator. Opportunities, registrations, and hours exist. |
| **Test Steps** | 1. Open the Program Overview page |
| **Expected Result** | The summary shows counts of opportunities, registrations, attendance, and total hours. No data changes. |
| **Actual Result** | (To be filled after test execution) |
| **Status** | (Pass/Fail/Blocked) |
| **Priority** | Medium |

### TC-034

| | |
|---|---|
| **Test Case ID** | TC-034 |
| **Related Requirement(s)** | FR-008 |
| **Title** | Program Summaries - No Data For Period |
| **Preconditions** | Logged in as an Administrator. No data exists for the selected period. |
| **Test Steps** | 1. Open Program Overview<br>2. Select a period with no activity |
| **Expected Result** | A clear "no information" message shows. |
| **Actual Result** | (To be filled after test execution) |
| **Status** | (Pass/Fail/Blocked) |
| **Priority** | Low |

### TC-035

| | |
|---|---|
| **Test Case ID** | TC-035 |
| **Related Requirement(s)** | FR-009 |
| **Title** | Notification - Opportunity Time or Location Changed |
| **Preconditions** | A volunteer is registered. Logged in as the Coordinator. |
| **Test Steps** | 1. Edit the opportunity time or location<br>2. Save<br>3. Log in as the registered volunteer and open notifications |
| **Expected Result** | The volunteer has a notification that names the opportunity and states what changed. |
| **Actual Result** | (To be filled after test execution) |
| **Status** | (Pass/Fail/Blocked) |
| **Priority** | Medium |

### TC-036

| | |
|---|---|
| **Test Case ID** | TC-036 |
| **Related Requirement(s)** | FR-009 |
| **Title** | Notification - Unregistered Volunteer Not Notified |
| **Preconditions** | A second volunteer is not registered for the opportunity. |
| **Test Steps** | 1. Edit the opportunity date and save<br>2. Log in as the unregistered volunteer |
| **Expected Result** | The unregistered volunteer receives no notification. |
| **Actual Result** | (To be filled after test execution) |
| **Status** | (Pass/Fail/Blocked) |
| **Priority** | Low |

### TC-037

| | |
|---|---|
| **Test Case ID** | TC-037 |
| **Related Requirement(s)** | NFR-001 |
| **Title** | Performance - Lists Load Within Two Seconds |
| **Preconditions** | Database holds hundreds of opportunities and thousands of records. |
| **Test Steps** | 1. Open Browse Opportunities<br>2. Open My Volunteering<br>3. Measure the time to display each page |
| **Expected Result** | Each page displays within two seconds. |
| **Actual Result** | (To be filled after test execution) |
| **Status** | (Pass/Fail/Blocked) |
| **Priority** | Medium |

### TC-038

| | |
|---|---|
| **Test Case ID** | TC-038 |
| **Related Requirement(s)** | NFR-003 |
| **Title** | Reliability - No Duplicate Records On Rapid Double Click |
| **Preconditions** | Logged in as a Volunteer. The opportunity has open spaces. |
| **Test Steps** | 1. Double-click "Register" quickly<br>2. Open My Volunteering |
| **Expected Result** | Only one registration exists. Remaining spaces fall by one only. |
| **Actual Result** | As expected. Automated test TC038 passed. |
| **Status** | Pass |
| **Priority** | High |

## 7. Code Repository and Branching Strategy

**Repository:** https://github.com/anwar7akim/volunteer-platform

- **main:** stable code. The team merges to main only at a tagged release.
- **development:** ongoing development. All feature branches merge here through a pull request.
- **Feature branches:** one branch for each key feature, for example `feature/auth`, `feature/opportunities`, `feature/registration`, `feature/coordinator-tools`. Each branch includes the unit tests for its feature.
- **Commits:** short messages in the Conventional Commits style, for example `feat(auth): add role-based login`.
- **Release tag:** the team tags the submitted version `ITR2.1`.

## 8. Task Allocation and Timeline

### 8.1 Task Breakdown

| Team member | Coding | Testing | Documentation |
|---|---|---|---|
| Anwar Bonunagui | Project setup, data storage layer, role-based login (US1) | Unit tests TC-001 to TC-007, TC-038 | Repository and branching section, test results |
| Mohammad Jalili | Browse and search (US2), register for opportunity (US3) | Unit tests TC-008 to TC-016 | Traceability matrix |
| Jana Hussien | Post, edit, and cancel opportunities (US6), attendance and hours (US7) | Unit tests TC-023 to TC-030 | Prototype overview, updated use cases |
| Crystal Mumtaz-Shah | Withdraw (US4), My Volunteering (US5), admin screens (US8), notifications (US9) | Unit tests TC-017 to TC-022, TC-031 to TC-037 | Test plan, sprint plan updates |

The team must confirm this allocation. Each member also reviews one pull request from another member.

### 8.2 Sprint Plan

| Week | Plan |
|---|---|
| Week 1 | Set up the Java project, the data storage layer, and the test framework. Implement login and role checks, browse and search, registration, and opportunity posting. Write unit tests for each. |
| Week 2 | Implement withdraw, My Volunteering, attendance and hours, admin screens, and notifications. Refine Week 1 features. Run all test cases. Record Actual Result and Status. Update the coverage status. |
| Week 3 | Fix defects found in testing. Finalize the prototype. Gather feedback from the instructor and teammates. Tag `ITR2.1` and submit the PDF to eClass. |

## 9. Submission Checklist

- Push all code and documents to GitHub.
- Tag the version `ITR2.1`.
- Submit this PDF on eClass: prototype overview, updated requirements and use cases, test plan and results, updated iteration plan.